

GABRIEL GALLO

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EDUCATION

University of California, Berkeley

B.S. in Electrical Engineering and Computer Science

Dec 2026

GPA: 3.66/4.00

Relevant Coursework: Computer Security, Data Structures and Algorithms, Digital Design and Integrated Circuits, Discrete Mathematics and Probability, Introduction to Robotics, Introduction to Full-Stack Development

TECHNICAL SKILLS

Languages: Python, Java, C/C++, Go, JavaScript, TypeScript, SQL, SystemVerilog, Assembly, HTML/CSS

Security & Systems: Applied Cryptography (AES-CTR, RSA-OAEP, HMAC-SHA512), Threat Modeling, OAuth 2.0, Secure System Design, Computer Security (coursework)

Frameworks & Libraries: React.js, Node.js, Express.js, Django, Django REST Framework, ROS2, NumPy, MongoDB

Developer Tools & Platforms: Git, GitHub Copilot, Cursor, AWS, Firebase, Cloudflare, Netlify

EXPERIENCE

Software Engineer

May 2025 – Aug. 2025

UNIFIBD

San Jose, CA

- Led end-to-end development of a cross-platform mobile application supporting individuals with Crohn's disease, reaching over 5,000 users.
- Migrated a legacy Swift iOS app to Flutter, consolidating iOS and Android into a single codebase and doubling platform reach.
- Published and maintained the app on the Apple App Store and Google Play Store, increasing accessibility across platforms.
- Reduced app load times by 30% through performance optimization, improving responsiveness and user retention.
- Collaborated with a team of 3 engineers to implement platform-specific enhancements and conduct cross-device testing, ensuring consistent UX across Android and iOS.

PROJECTS

Flow | *Angular, TypeScript, SCSS, Django, PostgreSQL, AWS, Gemini API*

- Built an authenticated productivity dashboard unifying email and calendar with collapsible sidebar navigation, using Angular, TypeScript, and SCSS.
- Implemented secure authentication via Google OAuth 2.0 (django-allauth) and built AI-powered email summarization, categorization, and event detection using NLP (spaCy, NLTK) and the Gemini API.
- Architected the backend with Django, Django REST Framework, and PostgreSQL, using Celery and Redis for background task processing.

TurtleBot Interceptor | *Python, ROS2, OpenCV, YOLO, SLAM, NumPy*

- Developed a Python/OpenCV cone detector with color and shape filtering plus distance estimation, improving detection precision from 78% to 93% in varied indoor lighting.
- Engineered an MPC controller integrating an Extended Kalman Filter, Monte Carlo Localization, local grids, and cone detections, reducing intercept time by ~1.2s and cutting path oscillations by ~30%.
- Instrumented visualization for cones, MPC trajectories, and obstacles, cutting debugging time during field tests by ~25%.

RISC-V 3-Stage Pipelined CPU | *SystemVerilog, Xilinx Vivado, PYNQ (AUP-ZU3)*

- Designed and implemented a 3-stage pipelined RISC-V processor in SystemVerilog targeting a Zynq UltraScale+ FPGA.
- Resolved data and control hazards in the pipeline to achieve a target CPI < 1.2 at a clock frequency above 100 MHz.
- Integrated a UART interface for BIOS tethering and hardware-in-the-loop debugging using the Xilinx ILA.
- Implemented support for the Zephyr RTOS shell, demonstrating system-level integration of hardware and software.

Secure File Storage System | *Go, Cryptography (AES-CTR, RSA-OAEP, HMAC-SHA512), Integration Testing*

- Architected a decentralized, stateless file storage client in Go operating against a global network adversary, using AES-CTR, RSA-OAEP, and HMAC-SHA512 to guarantee data confidentiality and integrity on an untrusted remote datastore.
- Modeled attacker capabilities (eavesdropping, malicious server, tampering, replay) and designed cryptographic countermeasures accordingly, removing any single point of trust in the system.
- Designed a dynamic multi-user sharing hierarchy using a centralized Sharing Tree, decoupling file appends from metadata growth to satisfy a strict bandwidth-efficiency requirement.
- Authored integration and tampering test suites covering adversarial edge cases, including empty files, multi-device synchronization, and structural datastore corruption.